Mahmoud Ahmed Abdelghani

Computer Engineering & Systems Student | Flutter / Mobile Developer

Zagazig, Sharqia, Egypt +20 100 137 3691 mahmoudabdelgani346@gmail.com

LinkedIn GitHub Portfolio Codeforces

Professional Summary

Computer and Systems Engineering student at Zagazig University (Expected Graduation: 2027) focused on cross-platform mobile development with Flutter and Dart. Hands-on experience building mobile applications with BLoC/Cubit, Firebase, REST API integration, and responsive UI/UX. Strong foundation in data structures, algorithms, and competitive programming, with experience in Generative AI, LLMs, and prompt engineering.

Education

Zagazig University, Faculty of Engineering **Egypt**

Bachelor of Science in Computer Engineering and Systems (Cumulative GPA: Very Good) *Expected Graduation: 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Database Systems, Embedded Systems, Microprocessors, Control Systems, Digital Logic Design

Technical Skills

Programming: Dart, Java, C++, C, Python, JavaScript, HTML, CSS, Assembly (MIPS)
Mobile: Flutter, BLoC, Cubit, GetX, Firebase (Auth, Firestore, Cloud Messaging, Storage), Responsive UI/UX
Software Engineering: Clean Architecture, SOLID Principles, OOP, Design Patterns, Unit Testing, Git, GitHub
AI / ML: Generative AI Fundamentals, Large Language Models (LLMs), Prompt Engineering, NVIDIA AI Tools
Databases: SQL, MySQL, Firebase Firestore, NoSQL
Embedded / Hardware: Arduino, AVR Microcontrollers, Embedded Systems, Bluetooth Serial, Hardware-Software Integration
Tools & Desktop: Android Studio, VS Code, Figma, Linux, Bash, JavaFX, Dio, GetIt, Secure Storage

Featured Projects

Ma3refa Mobile | *Flutter, BLoC/Cubit, Dio, Secure Storage, Figma* **GitHub Link**

- Built an interactive AI-powered quiz and personalized learning application from scratch in 10 days.
- Implemented quiz flows and state management using BLoC/Cubit.
- Integrated Dio for API communication, GetIt for dependency injection, and Secure Storage for local persistence.
- Designed the complete UI/UX in Figma with a focus on usability and accessibility.

Sampark Chat App | *Flutter, GetX, Firebase (Auth, Firestore, Cloud Messaging)* **GitHub Link**

- Built a real-time messaging platform supporting one-to-one and group chats with multimedia sharing.
- Implemented authentication and group-chat features using Firebase Authentication and Firestore.
- Integrated Cloud Messaging and Storage and used GetX for state management.

Laza - E-Commerce App | *Flutter, BLoC/Cubit, Firebase* **GitHub Link**

- Developed a full-featured e-commerce mobile application with product browsing, cart, wishlist, reviews, and ratings.
- Integrated Firebase Authentication and Cloud Firestore and implemented local caching for improved performance.
- Built responsive mobile interfaces with Flutter and BLoC/Cubit.

Laser Setter | *Flutter, C++, Arduino, Bluetooth* **GitHub Link**

- Created a Flutter control interface for a hardware-based laser leveling system.
- Sent real-time angular data to servo motors through Bluetooth serial communication.
- Developed Arduino firmware in C++ for hardware control and sensor integration.

File System Simulation | *Java, JavaFX, Trees, Linked Lists* **GitHub Link**

- Simulated file-system structures and operations using Trees and Linked Lists.
- Implemented file creation, deletion, navigation, and metadata management with Java, JavaFX, and Gson.

Training & Experience

Digital Egypt Pioneers Initiative (DEPI)

Mobile Development Trainee (Flutter)

- Completed training covering Dart, Flutter, Firebase, Git/GitHub, state management, responsive UI/UX, and testing.
- Built mobile applications and a capstone project using Flutter and Firebase.

ITI Zagazig & NVIDIA Mentorship Program

Generative AI & Mobile Development Trainee

- Selected for a specialized Generative AI program covering LLMs, prompt engineering, NVIDIA AI tools, and AI fundamentals.
- Developed Flutter applications and practiced navigation, UI components, state management, and Firebase integration.

IEEE Brand Ambassadors Program

Mobile Development Mentor

- Mentored students in Flutter and Dart through workshops, technical guidance, and code reviews.

ICPC Zagazig Community

Competitive Programming Trainee

- Completed Level 0 and Level 1 competitive programming training focused on algorithms, data structures, and problem solving.
- Ranked 7th in Level 0 and 4th in Level 1 competitions; solved 533+ Codeforces problems with a maximum streak of 20 days.

Additional Information

Hackathons: NASA Space Apps Challenge Cairo (Developed a weather application using NASA APIs)
Student Activities: Mobile Track Member at Zigzag Student Activity and Zag Systems
Languages: Arabic (Native), English (Professional Working Proficiency)
Military Status: Postponed